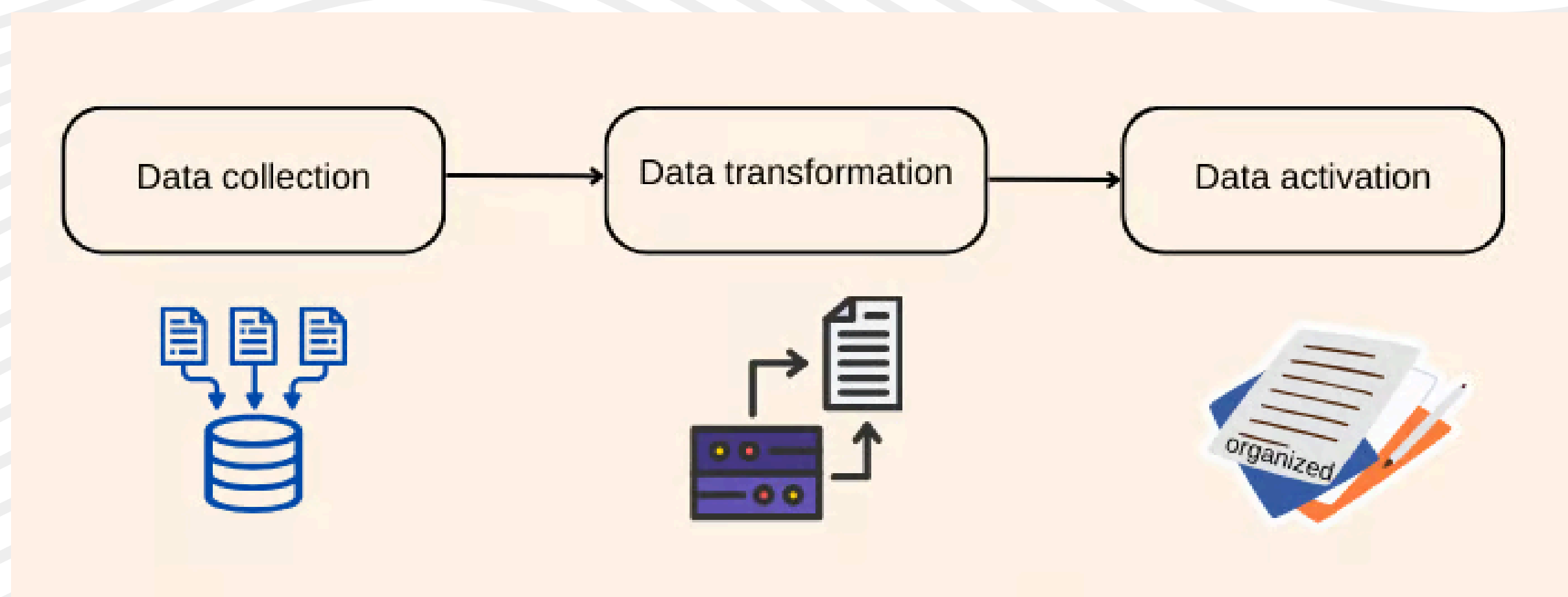


DATA ORCHESTRATION EXPLAINED: THE BACKBONE OF MODERN DATA PIPELINES



BY MICHAEL FERNANDES

WHAT IS DATA ORCHESTRATION?

Data orchestration is the process of coordinating, scheduling, and managing data workflows across multiple systems.

Without orchestration, workflows become difficult to manage and maintain.

Think of it as a conductor of an orchestra:

- Data sources = musicians
- Pipelines = instruments
- Orchestrator = conductor



shutterstock.com · 1952400367

WHY DO WE NEED DATA ORCHESTRATION?

Teams manually triggering workflows

Hundreds of ETL jobs running independently

Data arriving late or incomplete

Failed jobs going unnoticed

Difficult troubleshooting

No visibility into execution status

Business Impact

Delayed Dashboards

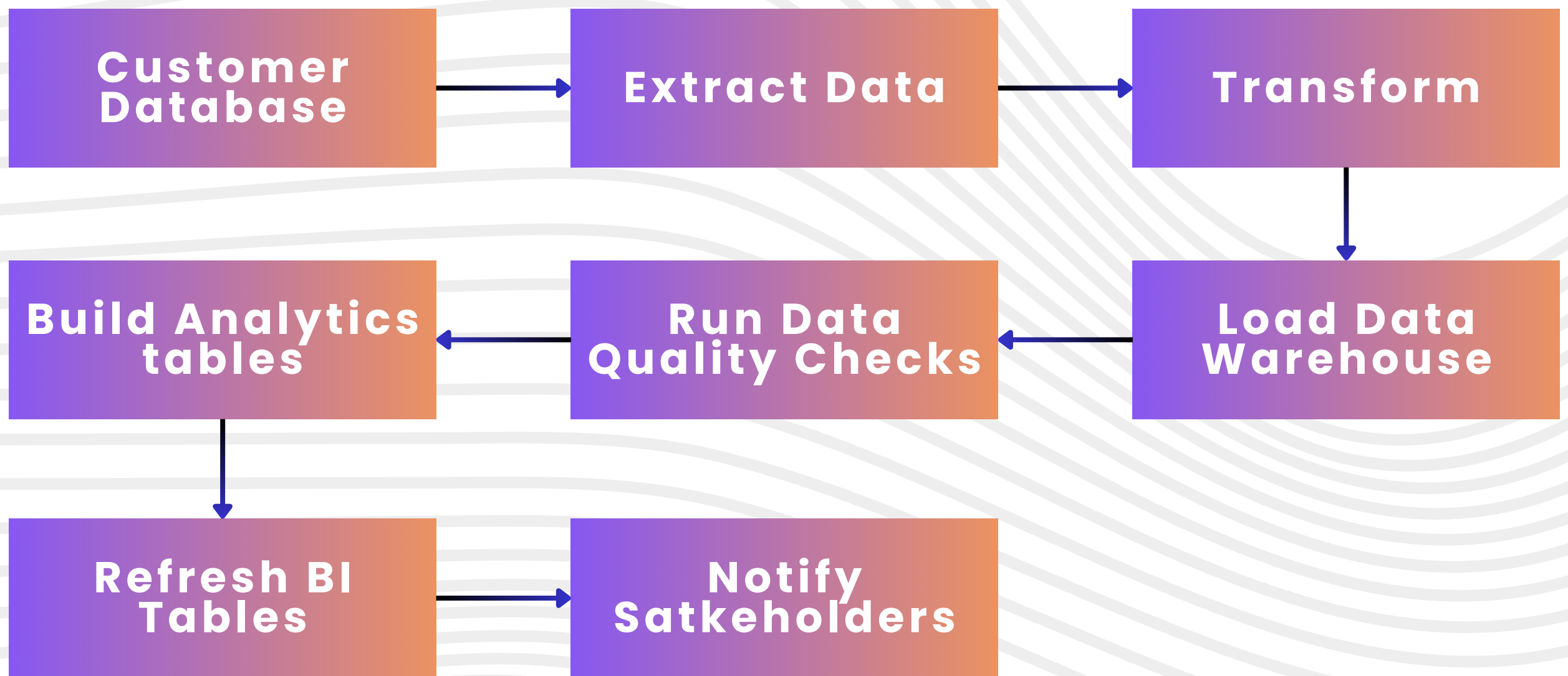
Incorrect reporting

Poor customer experiences

Slower Decision making

Increased Operational Cost

HOW DATA ORCHESTRATION WORKS



What Data Orchestration Does

Determines task order

Executes tasks automatically

Tracks dependencies

Retries failed tasks

Generates logs and alerts

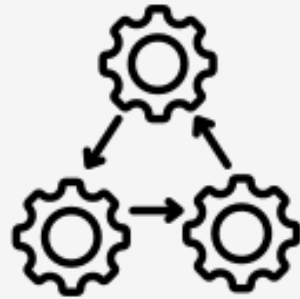
Provides workflow visibility

CORE COMPONENTS OF DATA ORCHESTRATION



Scheduling

Run workflows hourly, daily, weekly, or based on custom triggers.



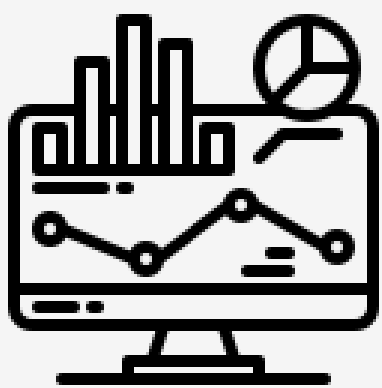
Dependencies

Ensure tasks execute in the correct order.



Workflow Engine

Executes and manages pipeline tasks.



Monitoring

Tracks execution status and runtime metrics.



Alerting

Sends notifications when failures occur.



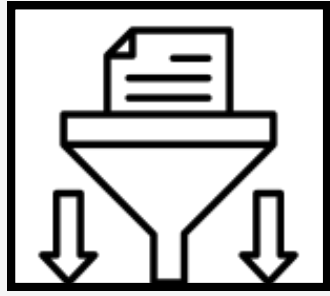
Logging

Stores execution history for debugging and auditing.

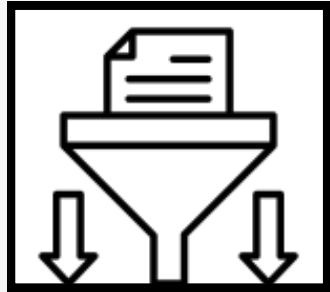
POPULAR DATA ORCHESTRATION TOOLS



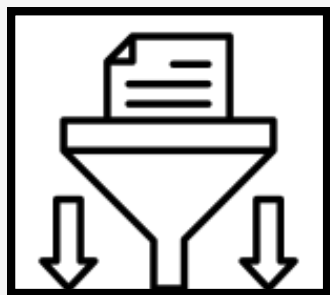
AIRFLOW WORKFLOW EXAMPLE



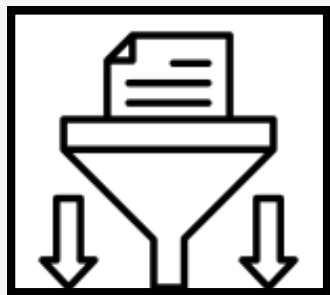
Task 1:
Extract Sales Data



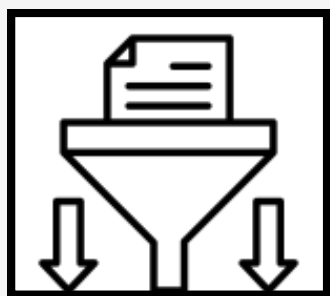
Task 2:
Clean & Validate Records



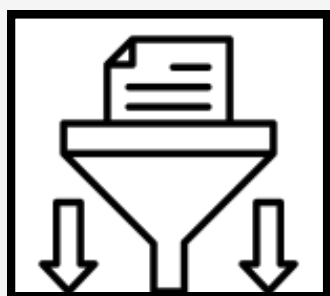
Task 3:
Apply Business Transformations



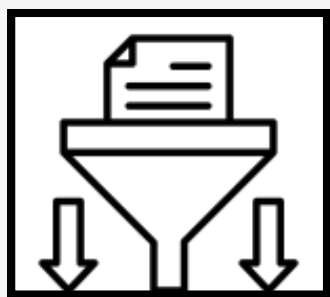
Task 4:
Load Data Warehouse



Task 5:
Run Data Quality Tests

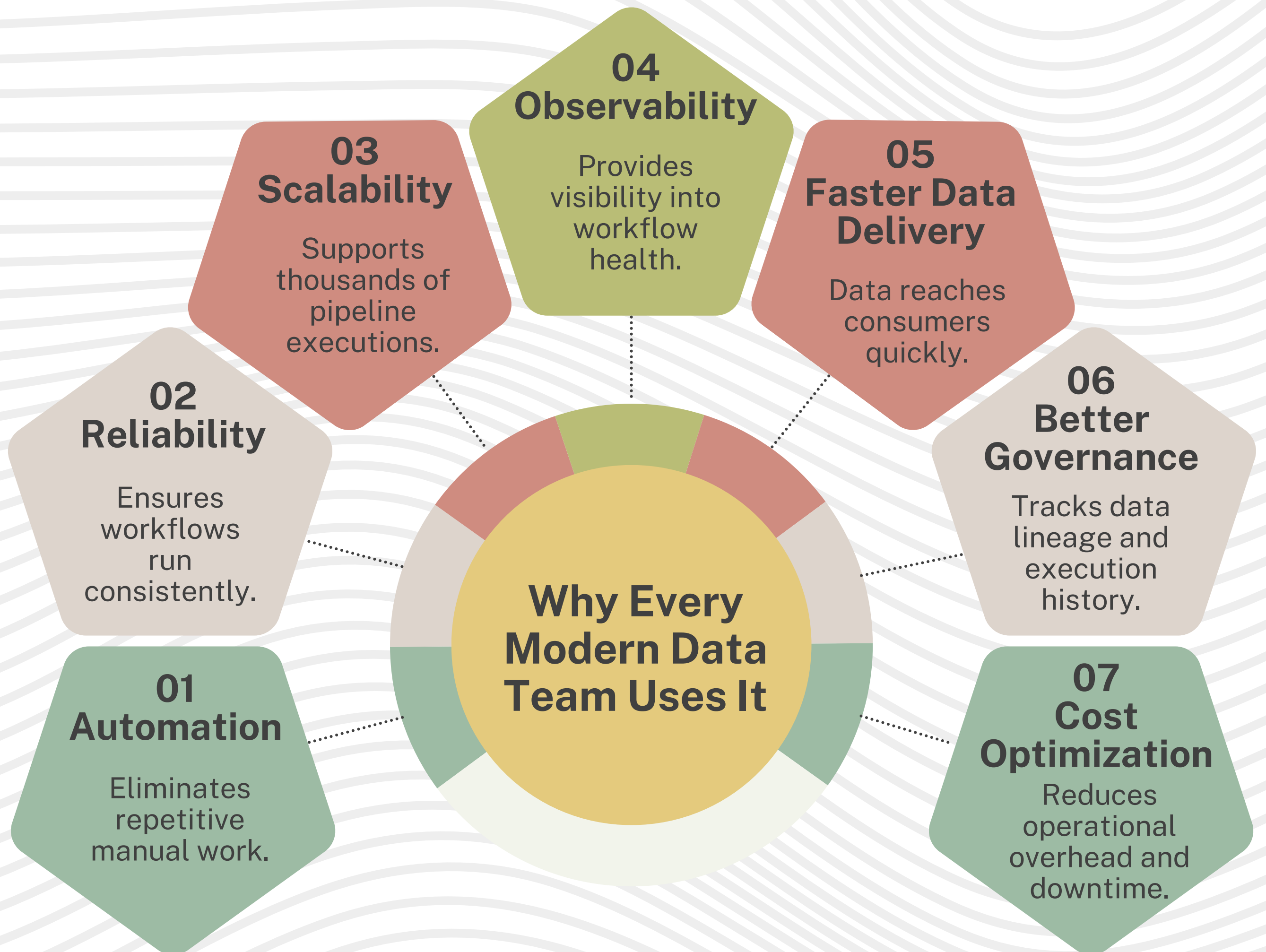


Task 6:
Generate Executive Reports

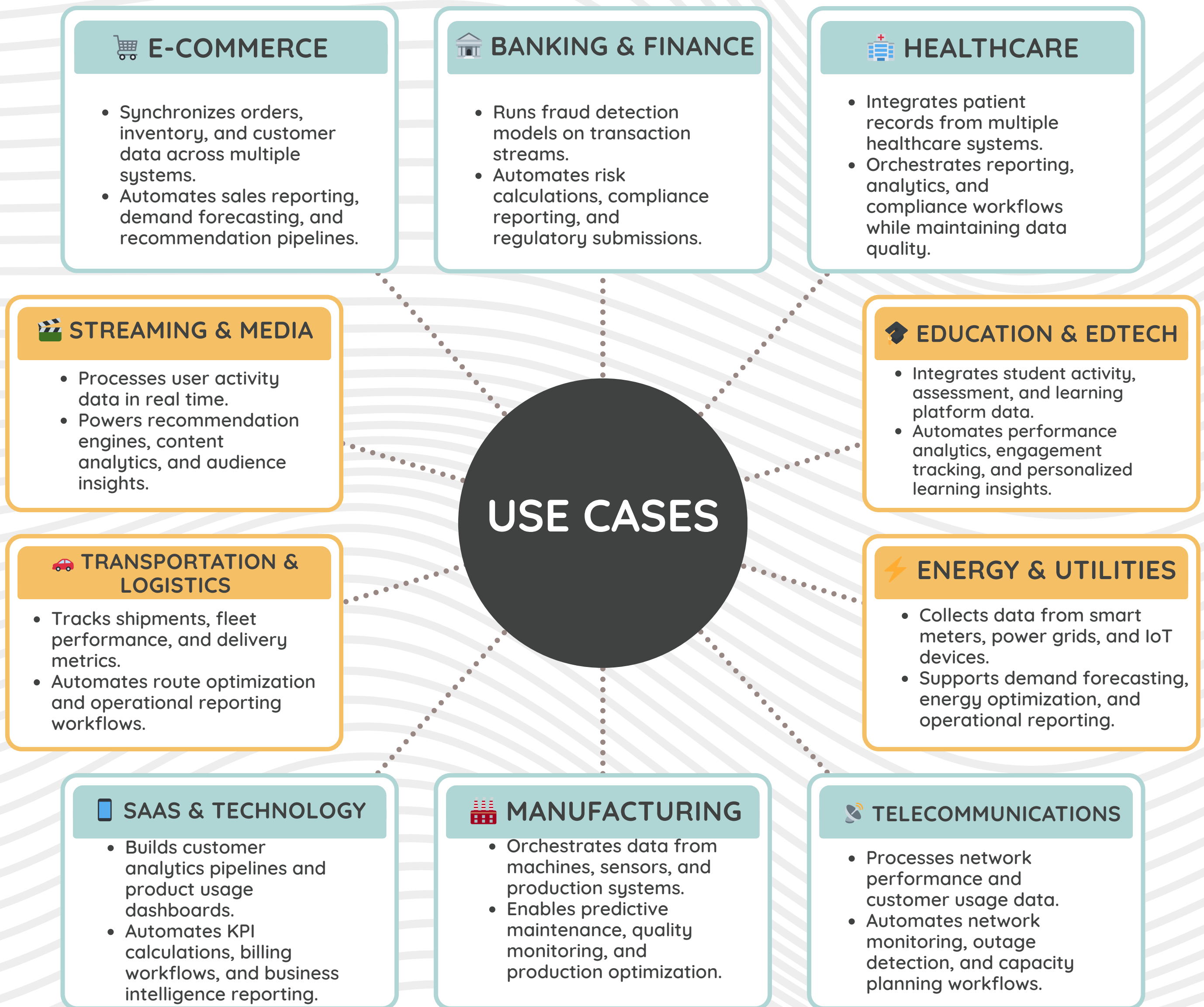


Task 7:
Send Success Notification

BENEFITS OF DATA ORCHESTRATION



REAL WORLD APPLICATIONS



WHAT EVERY DATA ENGINEER SHOULD REMEMBER

1. DATA ORCHESTRATION COORDINATES AND AUTOMATES DATA WORKFLOWS

2.IT MANAGES SCHEDULING, DEPENDENCIES, RETRIES, AND MONITORING

3.IT IMPROVES RELIABILITY, SCALABILITY, AND OBSERVABILITY

4.POPULAR TOOLS INCLUDE AIRFLOW, PREFECT, DAGSTER, AND CLOUD-NATIVE SOLUTIONS

5.DATA ORCHESTRATION IS A FOUNDATIONAL SKILL FOR MODERN DATA ENGINEERING



THANK YOU

BY MICHAEL FERNANDES